

Total No. of Pages

Roll No.

SECOND SEMESTER

M.Tech. (AI)

MID SEMESTER EXAMINATION

March, 2024

AI5402 NATURAL LANGUAGE PROCESSING

Time: 1:30 Hours

Max. Marks: 20

Note: Answer **ALL** questions.
Assume suitable missing data, if any.
CO# is course outcome(s) related to the question.

- 1[a] Consider the following corpus with positive and negative sentences. Create vocabulary from the corpus (ignore punctuation, upper, lower case) and feature vectors for each sentence using positive and negative frequencies of words (i.e. how many times a words appears in a class).

[3] [CO1]

Sentence	Sentiment
Loved the movie, very joyful.	Positive
Movie was dull, disappointing movie.	Negative
Movie thrilled me, loved it	Positive

- [b] Consider the following tweet and perform cleaning by removing stop words, punctuations, hashtag, handles etc. Perform stemming and lowering.

[1+2] [CO1]

Thrilled by game today! Energy high. #Joy #Victory

Disappointed by event. Long waits. #Frustrated

Band's performance fantastic! Loved it. #Happy #Loved

2. Consider the dataset in 1[a] again, performing sentiment analysis on a new sentence "Loved movie, but it was dull." using the Naive Bayes classifier. Use Laplace smoothing, if required.

[4] [CO2]

3. A convolutional neural network (CNN) has the following architecture:
Input layer: 3x32x32 image; *Convolutional layer1:* with 16 filters, each with size 5x5, using stride 1 and no padding; *Activation Layer1:* ReLU activation, *Max pooling layer1* with size 2x2 and a stride of 2;

Convolutional layer2 with 32 filters, each with size 3×3 , using stride 2 and no padding; *Activation Layer2*: ReLU activation, *Max pooling layer2* with size 2×2 ; *Fully Connected Layer1* with 120 neurons, *Activation Layer3* ReLU activation, *Fully connected layer2* with 84 neurons; *Output layer* with 10 neurons.

- [a] Find the size feature maps before *Fully Connected Layer1*. Show the calculation for each step. [3] [CO4]
 [b] Compute the number of learnable parameters in each layer of the above CNN architecture. [3] [CO4]

4 A Recurrent Neural Network (RNN) designed to predict the next number in a sequence. The RNN features an input layer for processing single numerical values, a hidden layer with two neurons utilizing the ReLU activation function, and an output layer for prediction. Given the sequence [2, 3, 4], and the model parameters—initial hidden state h_i , input to hidden weights W_{hi} , hidden to hidden weights W_{hh} , hidden to output weights W_{oh} , and biases for both layers set to 0. Calculate the hidden states and output at each timestep to predict the next value in the sequence. [4] [CO4]

$$\text{Given: } h_t = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, W_{hi} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, W_{hh} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, W_{oh} = \begin{bmatrix} 2 & -2 \end{bmatrix}$$

---Best of Luck---

END TERM EXAMINATION May, 2024

AI5402 NATURAL LANGUAGE PROCESSING

Time: 3:00 Hours

Max. Marks: 40

Note: Answer **ALL** questions.
Assume suitable missing data, if any.
CO# is course outcome(s) related to the question.

- ✓ 1.[a] Consider the following corpus of tweets regarding a review of some restorant. Perform tokenization, cleaning (punctuation, special characters removal, and lowering), stopword removal, and stemming. Create vocabulary from the corpus. [4] [CO1]

Review	Sentiment
Food was delicious, truly delicious! Service was amazing. Ambiance amazing. #delicious #amazing	Positive
Terrible experience! Food was cold, service cold and disappointing. #terrible #cold	Negative
Amazing food, delicious ambiance. Service amazing. #amazing #delicious	Positive
Food quality terrible, service disappointing. Cold and terrible experience. #terrible #disappointing	Negative

- [b] Consider vocabulary created in 1[a] and obtain vector representations for each tweets using Bag of Words (BoW) approach. List two major disadvantages of BoW. [2+2] [CO1, CO3]

- ✓ 2.[a] Suppose, you're using Naive Bayes for sentiment analysis on movie reviews. In your training dataset of 800 reviews (500 positive, 300 negative), the word "awesome" appears in 80% of positive reviews and 10% of negative reviews, while "dull" appears in 5% of positive reviews and 20% of negative reviews. Now, you are given a new review "The film was awesome but the pacing was dull.". Find the polarity (negative or positive) of the above review? 0.3 > 0.2 [4] [CO2]

- ✓ [b] Consider a sentiment analysis model that uses logistic regression to classify tweets as positive or negative. The model is trained with coffients $w_0 = 0.1$, $w_1 = 0.5$, and $w_2 = 0.6$. Consider a new tweet in retorant reviews: "Delicious food but disappointing service.". Let the 0.45

tweet is vectorized as $[1, x_1, x_2]$, x_1 and x_2 are sum of positive and frequencies (in corpus given in 1[a]) of each in the tweet. Find the polarity of the above tweet using the said logistic regression model.

[4] [CO2]

- 3.[a] Given a text dataset consisting of the following three sentences: "The cat sat on the mat."; "The dog sat on the log."; "The cat chased the rat."

Calculate the probability of the word that follows "cat" based on a bi-gram model. Assume all words are treated as lowercase, and punctuation is ignored for simplicity. Identify which word is most likely to be predicted after "cat" according to the bi-gram model probabilities.

[4] [CO3]

- [b] Suppose you have a corpus containing the sentence "Sunshine brightens the mood and warms the earth", and you are training a skip-gram model for word embeddings with a window size of 2. The current target word is "mood". What are the context words for "mood" according to the window size specified? If the vocabulary size of the model is 10, and the dimension of the embeddings is 4, how many parameters does the model need to learn?

[2+2] [CO3]

4. Answer any TWO of the followings.

- [a] Consider a simple RNN designed for language modeling. The vocabulary consists of three words: {START, hello, END}, encoded as one-hot vectors. The RNN has one hidden layer with two neurons. The initial hidden state $h = [0, 0]^T$ and the sequence input to the RNN is 'START', followed by 'hello'. The bias for the hidden layer and output are $[0, 0]^T$ and $[0, 0, 0]^T$ respectively. The weight matrices are given as:

$$W_{hi} = \begin{bmatrix} 0.5 & -0.5 & 0 \\ 0.3 & 0 & -0.3 \end{bmatrix}, W_{hh} = \begin{bmatrix} 0.1 & 0 \\ 0 & 0.2 \end{bmatrix}, W_{oh} = \begin{bmatrix} 0.1 & -0.1 \\ -0.1 & 0.1 \\ 0 & 0 \end{bmatrix}$$

Calculate the output probabilities for each word in the vocabulary after processing 'START'.

[4] [CO4]

- [b] Consider an LSTM cell with an input vector of size 2, a hidden state vector of size 2, and a current input of $[0.5, -0.5]$. Assume the forget gate, input gate, and output gate biases are all set to zero for simplicity, and the corresponding weight matrices for forget gate, input gate and cell state are given as below:

$$W_f = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0.2 & 0 & 0 \end{bmatrix}, W_i = \begin{bmatrix} 0.5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, W_c = \begin{bmatrix} 0.2 & 0 & 0 & 0 \\ 0 & -0.1 & 0 & 0 \end{bmatrix},$$

Given the that the initial hidden state h_{t-1} and cell state c_{t-1} both are zero vectors, find the update for the cell state c_t for this timestep. u. 54

[4] [CO4]

- [c] Explain the role of the forget gate in an LSTM cell and discuss how it contributes to the model's ability to avoid the vanishing gradient problem. Illustrate your explanation with a mathematical representation of the forget gate's function.

[4] [CO4]

5. Answer any TWO of the followings.

- [a] Consider a Transformer model is being trained on a corpus with minibatch of size 1. Let a sentence "The cat sat" is being processed in self-attention layer. The initial embeddings for tokens are: $The = [1, 0]$, $cat = [0, 1]$, $sat = [1, 1]$. The transformation matrices are given as $W^Q = W^K = W^V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. Calculate the query, key, and value vectors for each token. Also, compute the attention scores for each pair of query and key. K
- [b] Consider the sentence in 5[a] again, compute the final output vector of first attention layer for the token "sat" by applying the attention weights to the value vectors. Also, explain the significance of the values obtained.
- [c] Compare and contrast the Transformer architecture with recurrent neural networks (RNN). Also, explain the need of the positional encoding in Transformers.

[4] [CO4]

[4] [CO4, CO5]

[4] [CO5]

---Best of Luck---